

Lecture 14: Causal Inference

From “What Happened?” to “What Would Happen If. . .?”

Previously, on Lecture 13...

GLMs: Random component + systematic component + link function.

OLS as GLM: Normal + Identity. **Logistic:** Bernoulli + Logit. **Poisson:** Poisson + Log.

IRLS: Iteratively reweighted least squares for MLE in GLMs.

Deviance & AIC: Model comparison tools that generalize RSS and R^2_{adj} .

Today: All of L1–L13 answered “*what happened?*” and “*is the pattern real?*”

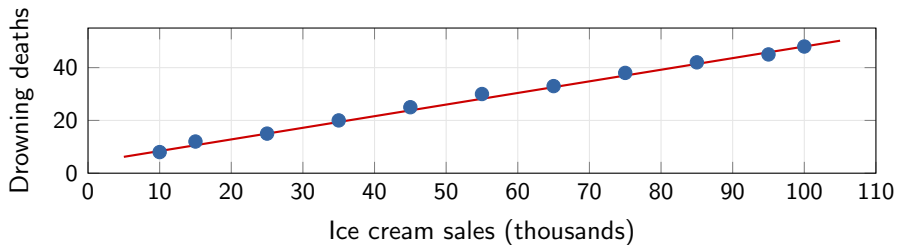
Now the hardest question: “**Did X cause Y?**”

Outline

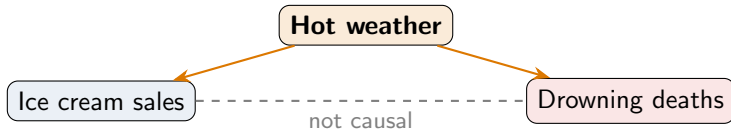
Part I: Correlation $\neq$ Causation

The most important sentence in statistics

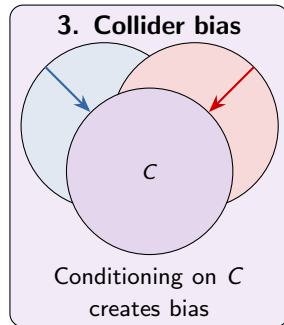
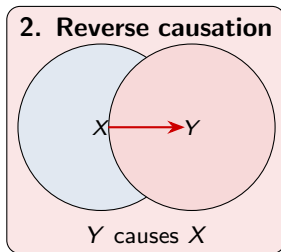
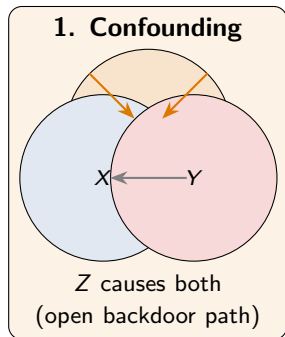
Spurious Correlations



$r = 0.95, p < 0.001$. Banning ice cream saves lives? **No.** A **confounder** (hot weather) drives both.



Three Reasons Correlation $\neq$ Causation

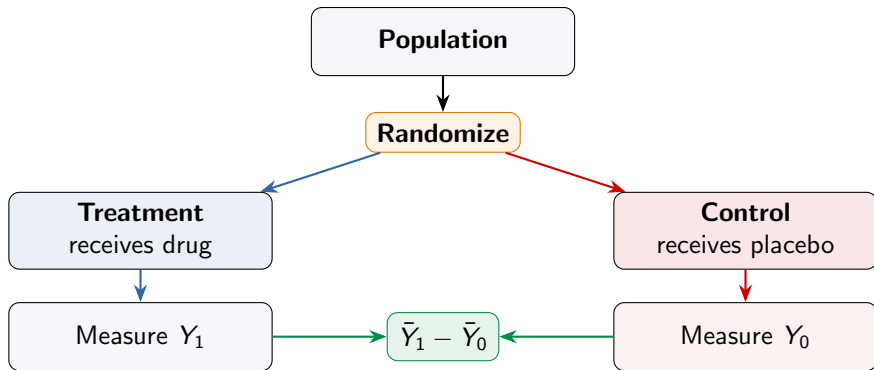


Regression, correlation, p -values — none can distinguish these scenarios.
We need **additional assumptions** or **experimental design** to claim causation.

Part II: The Gold Standard

Randomized controlled trials (RCTs)

Why Randomization Works



Randomization makes groups **comparable on average** — both observed and *unobserved* confounders. Any difference in outcomes is attributable to the treatment.

From A/B Tests to RCTs

Connection to L11

	A/B Test (L11)	RCT (clinical trial)
Setting	Website / app	Hospital / field
Units	Users / sessions	Patients / subjects
Treatment	New button color	New drug
Control	Old design	Placebo
Randomization	Cookie / user ID	Allocation sequence
Blinding	Users unaware	Double-blind ideal
Analysis	z-test / χ^2	t-test / survival analysis

A/B testing *is* a randomized experiment.
The statistical logic is identical — only the domain differs.

Part III: Potential Outcomes

The Rubin causal model

The Fundamental Problem of Causal Inference

Patient	$Y_i(1)$ (drug)	$Y_i(0)$ (placebo)	Causal effect $Y_i(1) - Y_i(0)$
Alice	120	?	?
Bob	?	140	?
Carol	115	?	?
David	?	135	?

The fundamental problem: Each person is either treated or not — never both. We can *never* observe the individual causal effect $Y_i(1) - Y_i(0)$.

The missing “?” entries are **counter-factuals**: what *would have happened*.

Notation: $Y_i(1)$ = outcome under treatment, $Y_i(0)$ = outcome under control. Only one is observed.

SUTVA: each person's outcome depends only on *their own* treatment (no interference).

Average Treatment Effect (ATE)

$$ATE = E[Y(1) - Y(0)] = E[Y(1)] - E[Y(0)]$$

We can't observe individual effects, but
we *can* estimate the **average** effect.

Under randomization:

$$E[Y(1) \mid T=1] = E[Y(1)]$$

$$E[Y(0) \mid T=0] = E[Y(0)]$$

Treatment is independent of potential outcomes.

So we can estimate ATE:

$$\widehat{ATE} = \bar{Y}_{\text{treated}} - \bar{Y}_{\text{control}}$$

The simple difference in means is **unbiased** for the ATE.

Other estimands: ATT (on the treated), CATE (conditional), LATE (local) — beyond our scope.

Why Randomization Identifies ATE

Step 1: We want $E[Y(1)] - E[Y(0)]$, but can only observe $Y(1)$ for treated and $Y(0)$ for control.

Step 2: Naïve comparison: $E[Y \mid T=1] - E[Y \mid T=0]$.

Step 3: This equals $\underbrace{E[Y(1) - Y(0) \mid T=1]}_{\text{ATT}} + \underbrace{E[Y(0) \mid T=1] - E[Y(0) \mid T=0]}_{\text{ATT}}$.

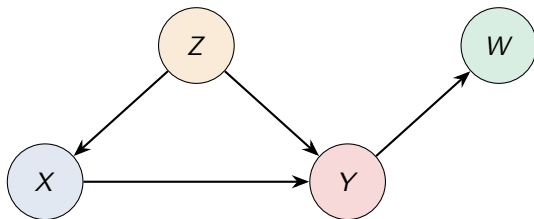
Step 4: Randomization ensures $T \perp\!\!\!\perp (Y(1), Y(0))$, so selection bias = 0 and ATT = ATE.

Without randomization: selection bias $\neq 0$. Healthier patients may choose the new drug. Richer users may see the new feature.

Part IV: Causal Graphs

DAGs: seeing the causal structure

Directed Acyclic Graphs (DAGs)



What is a DAG?

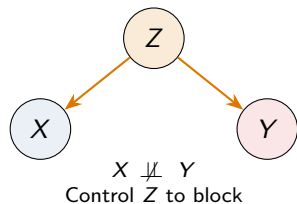
- Nodes = variables
- Arrows = direct causal effects
- “Directed”: arrows have direction
- “Acyclic”: no cycles ($X \rightarrow Y \rightarrow X$)

Reading the DAG:

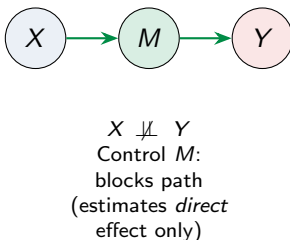
- Z causes both X and Y (confounder)
- X also directly causes Y
- W is a consequence of Y
- No arrow = no direct effect

Three Fundamental Structures

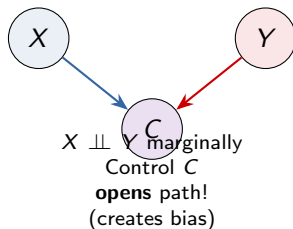
Fork (Confounder)



Chain (Mediator)



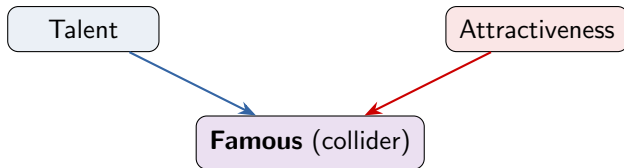
Collider



Key insight: Conditioning on a confounder *re-moves* bias. Conditioning on a collider *creates* bias. You must know the DAG to decide what to control for.

Collider Bias: A Concrete Example

The “talent vs attractiveness” paradox



In the general population:

Talent and attractiveness are independent. No correlation.

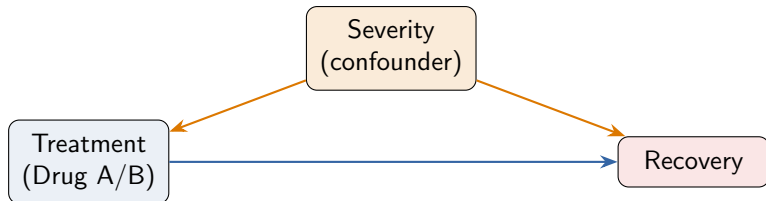
Among celebrities only:

Negative correlation! Very talented celebrities tend to be less attractive, and vice versa.

By conditioning on the collider (“among the famous”), we *created* a spurious association. Same logic: “Why are all my dates either boring or unstable?” — you’re conditioning on “agreed to date me.”

Simpson's Paradox Revisited

From L2 — now we have the language to explain it



	Drug A	Drug B	Who's better?
Overall	40%	50%	B
Mild	60%	55%	A
Severe	20%	15%	A

DAG tells us: Severity is a confounder $\Rightarrow$ **condition on it** (stratify).
Stratified result (A wins) is correct. Overall is misleading: A got more severe cases.

Part V: When You Can't Randomize

Observational causal inference

The Problem with Observational Data

RCT ✓

Randomization balances
all confounders
(observed + unobserved)

$$T \perp\!\!\!\perp (Y(1), Y(0))$$

Gold standard

Observational ✗

People self-select
into treatment

$$T \not\perp\!\!\!\perp (Y(1), Y(0))$$

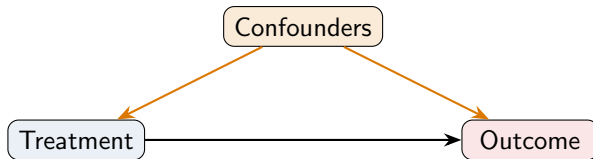
Need extra work

When can't we randomize? Ethics (can't assign smoking), cost, logistics, already-collected data.

We need methods that approximate what randomization does.

Strategy 1: Conditioning / Regression

Control for confounders directly



$$Y_i = \beta_0 + \beta_1 T_i + \beta_2 Z_{i1} + \beta_3 Z_{i2} + \dots + \varepsilon_i$$

$\hat{\beta}_1$ estimates the treatment effect **after adjusting for** $Z_1, Z_2, \dots$

Pros:

- Simple (just add covariates)
- Uses familiar tools (OLS, GLM)
- Can estimate heterogeneous effects

Cons:

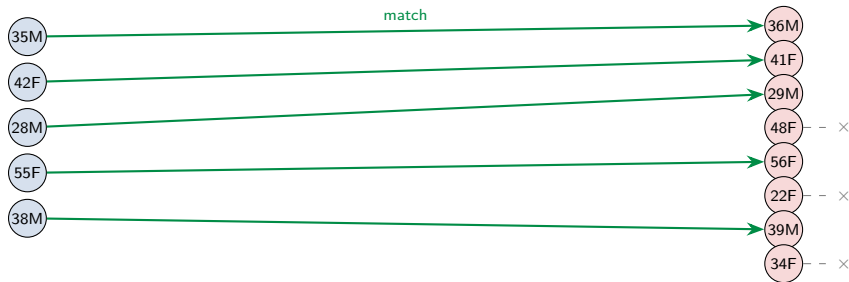
- Must know *all* confounders
- Assumes correct functional form
- Can't handle unmeasured confounders

Strategy 2: Matching

Find “twins” — one treated, one not

Treated

Control



For each treated unit, find a control with **similar co-variates** (age, sex, ...). Compare outcomes within matched pairs. Unmatched controls are discarded.

Strategy 3: Propensity Score

Rosenbaum & Rubin (1983)

$$e(X) = P(T = 1 \mid X)$$

The **propensity score** = probability of treatment given covariates.

Estimated by logistic regression: $\text{logit}(e) = X\beta$ (from L12/L13!)

PS Matching

Match on $e(X)$
instead of all X 's.
Solves dimensionality.

PS Weighting

Weight by
 $1/e$ (treated)
or $1/(1 - e)$ (control).
IPW estimator.

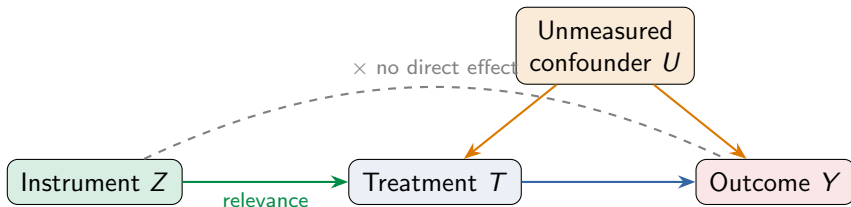
PS Stratification

Group by $e(X)$
quintiles. Es-
timate ATE
within each, average.

Key theorem: If treatment is ignorable given X , it's also ignorable given $e(X)$ alone.

Strategy 4: Instrumental Variables

When you can't measure all confounders



An **instrument** Z must satisfy:

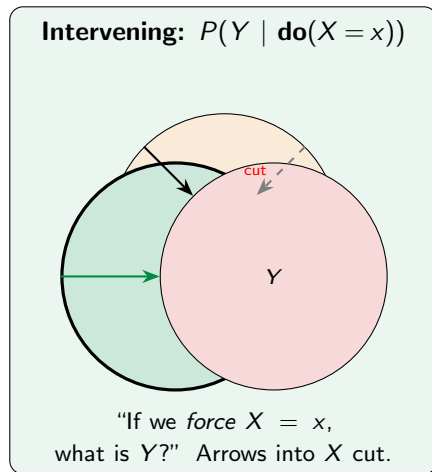
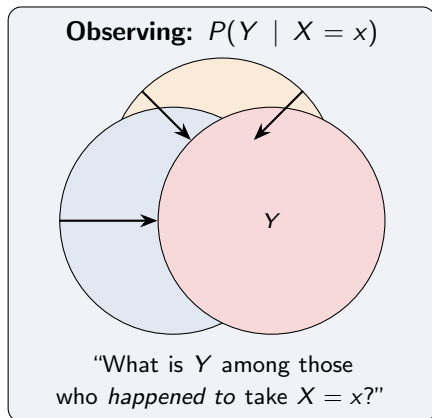
1. **Relevance:** Z affects treatment ($Z \rightarrow T$).
2. **Exclusion:** Z affects Y *only through* T (no direct $Z \rightarrow Y$ path).
3. **Independence:** Z is independent of unmeasured confounders.

Example: Effect of military service on earnings. Instrument: draft lottery number (random, affects service, no direct effect on earnings).

Part VI: Doing $\neq$ Seeing

$$P(Y \mid X) \text{ vs } P(Y \mid \text{do}(X))$$

Observing vs Intervening



The $\text{do}(\cdot)$ operator (Pearl) formalizes intervention: set X externally, severing all causes of X .

The Backdoor Criterion

When can we estimate causal effects from observational data?

A set of variables Z satisfies the **backdoor criterion** relative to (X, Y) if:

1. Z blocks every “backdoor path” from X to Y (paths with an arrow *into* X).
2. Z contains no descendant of X .

$$P(Y \mid \text{do}(X = x)) = \sum_z P(Y \mid X = x, Z = z) P(Z = z)$$

Adjustment formula: If Z satisfies the backdoor criterion, the causal effect is computable from observational data by conditioning on Z .

This is exactly what regression does when you “control for” confounders. The DAG tells you *which* variables to control for.

Part VII: In Practice

A checklist for causal claims

The Ladder of Causation

Pearl's three levels

Level 3: Counterfactual (Imagining)

“Would this patient have survived *if* they had taken Drug A instead of B?”

Tools: structural equation models, potential outcomes, twin networks

Level 2: Intervention (Doing)

“What *would happen* if we give everyone Drug A?” $P(Y \mid \text{do}(X))$

Tools: RCTs, adjustment formula, instrumental variables, propensity scores

Level 1: Association (Seeing)

“Among people who took Drug A, what fraction recovered?” $P(Y \mid X)$

Tools: regression, correlation, hypothesis tests — everything from L1–L13

increasing difficulty

Causal Inference Checklist

1. **State the causal question** clearly. What is the treatment? What is the outcome?
2. **Draw the DAG.** Identify confounders, mediators, colliders. Use domain knowledge.
3. **Can you randomize?** If yes $\rightarrow$ RCT / A/B test. You're done (L11).
4. **If not: identify a valid adjustment set** using the backdoor criterion.
5. **Choose method:** regression, matching, propensity scores, IV, diff-in-diff, ...
6. **Check assumptions.** Sensitivity analysis: how would an unmeasured confounder change results?
7. **Report honestly.** State assumptions. Acknowledge limitations. Avoid over-claiming.

Common Pitfalls

✗ Controlling for everything

Adding all available variables to regression can *introduce* bias (collider, mediator). Let the DAG guide you.

✗ Post-treatment conditioning

Controlling for a variable *caused by* treatment blocks part of the effect you're trying to measure.

✗ “Correlation = causation” with big data

More data doesn't fix confounding. $n = 10,000,000$ with selection bias is still wrong.

✗ Ignoring external validity

RCT on college students $\neq$ effect on elderly patients. The ATE depends on who's in the study.

Summary: Causal Inference

Correlation $\neq$ causation because of confounders, reverse causation, and collider bias.

Potential outcomes: $Y(1), Y(0)$. Fundamental problem = we only see one.
Goal: estimate ATE.

RCTs break confounding via randomization. A/B tests (L11) are RCTs.

DAGs encode causal assumptions. Three structures: fork, chain, collider. Back-door criterion.

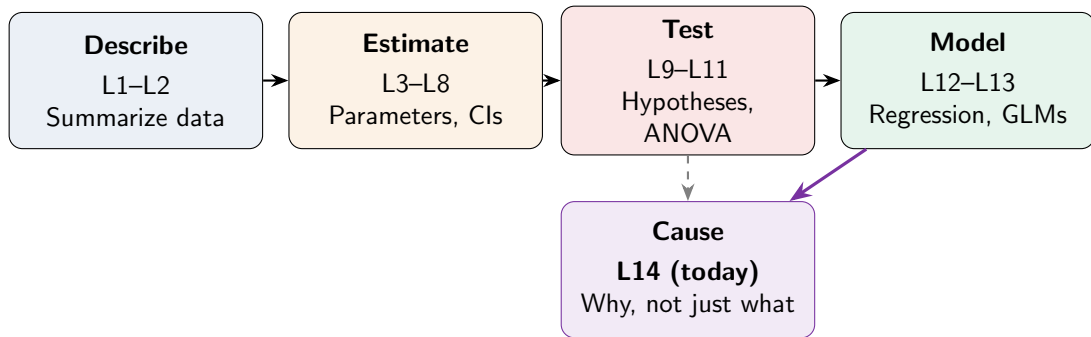
Observational methods: regression adjustment, matching, propensity scores, instruments.

do $\neq$ see: $P(Y | X) \neq P(Y | \text{do}(X))$ unless we adjust for the right variables.

No free lunch: every observational method requires assumptions. State them explicitly.

The Course So Far

From data to causation



Practical Exercise

1. **Draw DAGs:** Education $\rightarrow$ Income and Exercise $\rightarrow$ Weight loss. Identify confounders and mediators in each.
2. **Simpson's paradox:** Create a $2 \times 2 \times 2$ table where overall and stratified effects reverse. Verify with a DAG.
3. **Propensity scores:** Simulate confounded data. Compare naïve ATE, PS-matched ATE, and true ATE.
4. **Collider bias:** Simulate independent talent and attractiveness. Condition on “famous” (= high talent OR attractiveness). Show spurious correlation.

Homework

1. Surgery A has higher mortality than B in hospital data. (a) Draw a DAG with severity as confounder. (b) How would propensity scores estimate the causal effect? (c) What assumptions are needed?
2. Users who see more ads buy more ($r = 0.4$, $p < 0.001$). (a) List three alternative explanations. (b) Draw DAGs for each. (c) Design an A/B test (from L11) to answer the causal question.
3. Collider simulation: $X, Y \sim N(0, 1)$ independent, $C = X + Y + \varepsilon$. (a) Compute $\text{cor}(X, Y)$. (b) Compute $\text{cor}(X, Y \mid C > 2)$. Explain.
4. “Distance to hospital” as IV for “receiving surgery.” Check each IV assumption. Which is hardest to defend?

Recommended Resources

Hernán & Robins: *Causal Inference: What If* (2020) — free online. The modern standard.

Pearl, Glymour, & Jewell: *Causal Inference in Statistics: A Primer* (2016) — concise, DAG-focused.

Cunningham: *Causal Inference: The Mixtape* (2021) — free online, economics perspective.

Angrist & Pischke: *Mostly Harmless Econometrics* (2009) — IV, regression discontinuity, diff-in-diff.

Python: `dowhy` (Microsoft), `causalml` (Uber) — causal inference libraries.

Questions?

This concludes the Statistics lecture series.

L1–L14: From data description to causal inference.